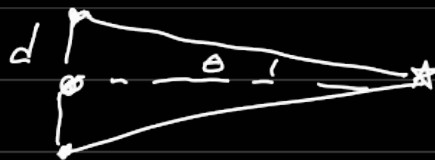


$$\theta = \frac{1 \text{ AU}}{d}$$



$$\frac{l_2}{l_1} = (100)^{\frac{1}{5}(m_1 - m_2)}$$

$$m_1 - m_2 = 2.5 \log_{10} \left(\frac{l_2}{l_1} \right)$$

apparent: m

absolute: M :

at 10 par.

$$l \sim \frac{1}{d^2}$$

$$\begin{aligned} m - M &= 2.5 \log \left(\frac{d^2}{10^4} \right) \\ &= 5 \log \left(\frac{d}{10} \right) \end{aligned}$$

$$m - M = 5 \log(d) - 5$$

* Stat. Mech.:

• look at an ensemble of particles

• Take averages

• Probabilistic:

part. function:

$$Z = \sum_n e^{-E_n/k_B T}$$

$$P(n) = e^{-E_n/k_B T} / Z$$

* Probability of a particle to have energy E_n .

why?: $\sum_n P(n) = 1$

• Averages: Notation: $\bar{E} = \langle E \rangle$

$$\langle \hat{H} \rangle = \langle E_n \rangle = \sum_n E_n e^{-E_n/k_B T}$$

etc

- Temp: Measure of motion/energy of system

- V_{rms} : Main value which gives us overall info about movement of gas particles

$$V_{rms} = \sqrt{\langle V^2 \rangle}$$

$$\langle V^2 \rangle = \sum_n v^2 e^{-E_n/k_B T} \text{ etc...}$$

- Take cont. limit

$$\sum_n \rightarrow \int d^3 n = \frac{V}{(2\pi)^3} \int d^3 k$$

* Equipartition Thm:

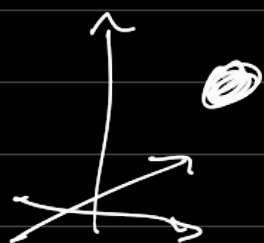
- For most simple non-rel. gases, the system follows the following relationship

$$\langle E_n \rangle = \frac{f}{2} k_B T \quad (\text{non-rel. gas})$$

f : D.O.F of the system
Rel. gas

$$\langle E_n \rangle_{rel} = f k_B T \quad (\text{Rel. gas})$$

- Monatomic gas:



- Can only move in (x, y, z)

$$f = 3$$

- Photon: tricky, uses $\frac{1}{2}$
 $f=2$ (polarizations)

* Application: ISM: (Gas & Dust)

- Hydrogen: $\langle E_n \rangle = \frac{3}{2} k_B T$

K.E.: $\frac{1}{2} m_H v_{rms}^2 = \frac{3}{2} k_B T$

$$v_{rms} = \sqrt{\frac{3 k_B T}{m_H}}$$

- We know, k_B, m_H : $\underbrace{\sqrt{\frac{3 k_B}{m_H}}}_{\tilde{c}} T^{1/2}$

$v_{rms} = \tilde{c} T^{1/2}$

- Use this to get a mental picture of how quickly gas is moving.

→ Helps me think about these things

* Dim analysis:

- Things like n_H : number density
 always think about as

$$n_H = \frac{N}{V} \leftarrow \begin{array}{l} \# \text{ of part. in a} \\ \text{Volume} \end{array}$$

- $S_x = \frac{N_x}{t}$

* ISM Not Covered:

(i) Dynamics: energetics & shocks

- Magnetic shocks

- Ad. shocks...

(ii) stellar winds / Supernovae

(iii) Rel. particle physics

(iv) Synchrotron radiation

(v) ...

(v) Hot plasmas

(vi) Pulsars & Radio jets

(vii) Many other ISM factors

